BuyerUnion | Implementation PRD v3

CalHacks 13.0 | Revised October 23, 2026 | Implementation handoff

1. Core Details and Why

Build: a marketplace for manufactured consumer products: buyers describe a need and constraints; merchant agents compete with permitted offers; a bounded council challenges claims; independent web evidence informs the final decision. Technology, household accessories, appliances and clothing share this flow. Laptops are a demonstration, not the product boundary. Nothing is bought or fulfilled in the real world during the demo.

Confirmed scope: four teammates, two own UI in Figma; no existing application; manufactured products, new/refurbished where applicable; U.S./USD; San Francisco delivery; buyer-selected deadline; disclosed mock merchant stock/terms; real web evidence; Visa sandbox attempted, labeled simulation otherwise. No language/framework preference. This supersedes both real-inventory-first and laptop-only assumptions.

Chosen defaults: TypeScript, React/Vite, Convex/Auth/Workflow, direct Gemini API, Apify; actual frontend hosted on Vultr. Three fictional merchants; one primary product type and up to two related product types per request, all from one merchant; at most five offer cards. Each line quantity is 1-10. Category profiles supply typed rules; adding categories must not require changing the council, checkout or screen sequence. No second database, real-money mode or multi-seller checkout.

Coverage boundary: the architecture accepts extensible product categories; launch profiles cover laptops/tech accessories, clothing, household accessories and small appliances. Do not claim universal verified inventory or evidence coverage. Unknown categories stay editable drafts until a validated profile exists. Services, consumables and regulated goods are outside this prototype; large-appliance installation is not promised.

Why: make requirements, commercial terms and evidence inspectable together, while reducing buyer comparison work and merchant quotation effort. Debate is useful only when it improves a valid offer or resolves a material claim. Longer reasoning is not evidence of a better outcome.

Visa criterion	Required demonstration
Innovation 25%	Tailored merchant offers and a challenge that produces a useful correction/concession; avoid claiming to invent RFQs.
UX 25%	Broad and precise input, editable requirements, concise progress, understandable offers and clear purchase approval.
Technical 20%	Actual scoped tool calls, enforced policies, real source retrieval, persisted state and recovery.
Commerce 15%	Measure comparison/quotation effort and fulfilled demo offers; no invented savings or conversion uplift.
Trust 15%	Mock disclosure, separate commercial/private data, applicable citations, truthful incomplete states and bounded approval.

Readiness boundary: this document resolves product and architecture choices sufficient to begin the demo. It does not certify provider access, scraping success, measured performance, or prize eligibility. Required source/API smoke tests are listed on page 10. Live merchant adoption and economics remain future validation questions, not prototype blockers.

Audit changes: category-based requirements replace laptop assumptions; mock inventory is authoritative; stack/auth, ranking, council bounds, evidence applicability, reservation/payment recovery and acceptance cases are explicit. Figma ownership and frontend/backend handoff are defined. More compute is permitted only for a material unresolved issue, not repetitive persuasion.

2. Flow: screens, requirements and comparison

Buyer screens: intake -> confirm requirements -> progress -> comparison/evidence -> final quote -> result. Preserve the request on reload. Show "Demo merchants and inventory; no real purchase" and evidence/payment mode badges. Merchant screens: own catalog, policy controls, pause toggle, own bids/rejections and permitted council events.

1. Establish an anonymous buyer session before saving. Accept up to 4,000 characters. Broad requests trigger material clarification; exact requests prefill the same editable form. Ask at most three grouped clarification questions, then use the form for unresolved required fields.

2. Confirm category/profile, quantity, all-in USD budget, deadline, San Francisco area, acceptable condition, must-haves and optional related products. Budget/deadline start unset. Show profile condition defaults and lowest-total priority; no optional items are selected silently. Resolve ambiguous categories before running.

3. Persist an immutable request version and start a run. Show stage, elapsed time, merchants responded and evidence coverage. Allow cancel before execution; edits cancel obsolete work and create a new version.

4. Accept valid offers, run council, research finalists. Show 3-5 offers only when that many qualify; otherwise show the true count. Expose itemized demo total, configuration/condition, arrival, seller warranty, returns and evidence gaps.

5. Buyer inspects sources and chooses. Revalidate/reserve only the selected bundle; show a three-minute server countdown, changed terms, and final approval. No inferred approval.

Shared fields and category profiles

Common constraints: category, brand/model if specified, quantities, condition, total ceiling, arrival, minimum seller warranty/returns. Product constraints are typed entries: line ID, profile field ID, allowed operator, value/unit, hard or soft. Pin the immutable profile version through validation/checkout; updates apply to new requests. Profiles define labels, types, ranges, units, operators, clarification, variant identity, compatibility and evidence scope. Models propose values; server validators enforce them.

Profile	Launch-specific rules
Laptop / tech accessory	OS, RAM/storage GB, weight grams, USB-C charging; refurbished battery-health percent; explicit supported-device compatibility.
Clothing	Garment type, brand-specific size label/system, color, material percentages, garment measurements in mm. No universal M-to-M equivalence; fit preference is advisory. Default new only.
Household accessory	Item type, dimensions mm, material, color, mounting/interface compatibility. Do not infer load capacity or attachment safety. Default new only.
Small appliance	Type, dimensions mm, capacity with named unit, voltage V/frequency Hz and plug/region. Compare measured noise/energy only with units and test basis. Default new/refurbished.

Operators: equals/in/min/max/containsAll, allowed per field; convert compatible units before comparing. Reject unknown fields/operators or contradictory bounds. Unknown hard facts fail; subjective claims such as "flattering" remain preferences. New profiles are operator-reviewed configuration, never generated executable rules.

Store UTC milliseconds; show America/Los_Angeles. Date-only means 23:59:59 local, confirmed explicitly; reject past dates. Demo arrival = quote time plus configured calendar days. No street address/card data. UI must support keyboard use, readable status announcements and preserved focus.

Deterministic ordering and refill

Filter hard requirements/private budget first. Default: total ascending, arrival ascending, warranty descending, stable offer ID. "Soonest" puts arrival first; "Warranty" puts warranty first, then total/arrival. Sentiment never changes this order. Compare complete requested bundles; explain category-specific tradeoffs separately.

At most two candidate offers per merchant. An accepted revision supersedes its parent in the current pool; retain history. Take the top offer per merchant where available, then fill remaining places by the selected order, maximum five. Different sellers may offer the same configuration; label it honestly. If an applicable factual contradiction invalidates a finalist, record why and promote the next valid reserve. Research each new identity once, one refill pass within run limits. Never relax constraints or restart bidding silently.

3. Catalog, mock commerce and seed requirements

Identity: Product = family/model; Variant = exact configuration, size/color/material or regional version; Listing = merchant's stock/condition/terms for that variant. Manufacturer part number/GTIN is optional and never invented. Include category/profile version and per-attribute provenance: manufacturer, real report or synthetic demo. Stock belongs to listings, not families.

Laptop pitch fixtures: internal IDs, not manufacturer part numbers. Sources establish specifications, never stock. [S7-S10]

ID	Canonical configuration	Source
P1	MacBook Air M2 2022; 8-core CPU/8-core GPU; 16 GB/512 GB	Apple US M2 [S7]
P2	MacBook Air 13-inch M3 2024; 8-core CPU/8-core GPU; 16 GB/512 GB	Apple US M3 [S8]
P3	MacBook Air 13-inch M4 2025; 10-core CPU/8-core GPU; 16 GB/512 GB	Apple US M4 [S9]
P4	Same M4 family; 10-core CPU/10-core GPU; 24 GB/512 GB	Separate configuration [S9]
P5	ThinkPad T14 Gen 5 Intel; Core Ultra 5 125U; 16 GB/512 GB; Windows 11 Pro English	US model 21ML0053US; withdrawn status does not imply current stock [S10]
P6	MacBook Air M2 2022; 8-core CPU/8-core GPU; 8 GB/256 GB	Low-spec fixture [S7]

Retain source/checked date. Unknown attributes stay null. Apple OS family is macOS; no unsupported OS-version, weight or benchmark claims.

Fixed commercial fixtures

Fictional USD dollars; store cents. N=new, R=refurbished. Entry = product / condition / quantity / list / floor.

Merchant	Laptop listings	All-category demo policy
Demo Merchant A	P1/R/1/620/580; P3/N/2/990/950; P5/R/1/690/650; P6/R/1/480/450	Shipping 15; arrival +3 calendar days; 12-month demo seller warranty; 14-day returns.
Demo Merchant B	P2/R/1/750/710; P3/N/1/1020/965; P5/N/1/850/810; P6/N/2/550/520	Shipping 0; arrival +1 day; 24-month demo seller warranty; 30-day returns.
Demo Merchant C	P1/R/1/610/610; P4/N/1/1090/1090; P5/R/1/680/680; P6/R/1/460/460	Shipping 0; arrival +2 days; 12-month demo seller warranty; 14-day returns. No price concession.

Refurbished listings are unique units. Synthetic health: A/P1=92%, A/P5=90%, B/P2=88%, C/P5=85%; others unknown. These are not certified diagnostics. Unknown fails a mandatory health threshold.

Each merchant: ten fictional mice at list 25, ten USB-C-to-USB-A data hubs at 35; A/B floors 15/20, C floors equal list. Simulated P1-P6 compatibility only; no charging/display claims. Included charger is a demo condition flag.

Required non-tech fixtures

All below are fictional brands/products with synthetic attributes, not web-reviewed real items. Each merchant stocks five new units of every variant; A/B floors = 90% of list, C floors = list. Same policy table applies. Reviews show unavailable. Replace/add a genuine identified non-tech item for the real-source smoke test; never relabel these fixtures as real.

Variants	Typed attributes and USD list prices
C1 / C2: Demo Tee	Brand DemoWear; size system Demo US unisex; navy. C1: M, cotton 100%, garment chest circumference 1000 mm/length 700 mm, 25. C2: L, cotton 60%/polyester 40%, 1080/730 mm, 30.
H1 / H2: Demo Shelf	Wood, white, freestanding. H1: width/depth/height 500/300/600 mm, 80. H2: 800/400/800 mm, 100. No load-rating assertion.
A1 / A2: Demo Kettle	1.5 L, 220/160/230 mm. A1: 120 V/60 Hz, NEMA 1-15 US plug, 45. A2: 230 V/50 Hz, CEE 7/16 plug, 40. No safety certification asserted.

All prices are tax-inclusive fictional prices: tax line zero, "included in demo price." Total = sum(quantity x permitted unit price) + shipping once per bundle. No fee/cross-line discount. Agents choose floor-to-list prices; warranty/returns/shipping remain policy constants.

4. Merchant agents and adaptive council contract

One internal Gemini context per merchant; server binds scope before tool calls. Validate arguments and structured final output. Preserve SDK function-call identity/signature parts. No recursive agents, arbitrary browsing, shell access, or model-selected merchant identity. [S3]

Role	Allowed operations	Cannot do
Intake	Propose fields, clarify, explain confirmed request	Confirm requirements or purchase for the user.
Merchant	searchOwnCatalog, readOwnInventory, readOwnPolicy, readApprovedSpecs, proposeOffer, respondToChallenge, reviseOffer, abstain	Edit stock/policies; see ceiling/rival floors; reserve/pay.
Moderator	Read shared requirements/permitted claims; challenge; classify supported/contradicted/unknown	Purchase, alter hard constraints, rank rhetoric, view floors.
Evidence	Extract allowlisted sources, assess applicability, attach claims	Change prices, stock, permissions or approval.

Offer: request/run/version; listing IDs/quantities; integer unit prices; policy/inventory versions; claims/source IDs; reason or abstention. Server recomputes total, arrival and terms, validates constraints and persists immutable accepted versions. One repair after rejection within caps, otherwise abstain. Private-constraint rejection is generic to sellers; no budget headroom.

Events: event ID, request/version, run ID, round, server-bound role/merchant, offer version, type, timestamp. Types: proposal, accepted, rejected, challenge, response, revision, abstained, closed. Challenge names a claim/concession. Responses cite approved sources or admit uncertainty; changed goods/prices require a validated new offer version.

Compute policy

Default one round: claims arrive with bids; moderator asks at most one targeted question per merchant; each answers/revises once; moderator summarizes changes. Buyer gets a summary and optional claim/tool-outcome transcript, not hidden reasoning. Skip council with fewer than two qualifying merchants.

Allow one additional round only for a named material issue, a specific new tool result or permitted concession able to resolve it, and time/call/quota headroom. Log trigger. Stop on resolution, no material change, two rounds, cancellation or a limit. Prefer extra compute for relevant evidence checks over repeated persuasion. Defaults are not a proven cost-quality optimum.

Server-configurable caps: three concurrent merchants; two proposals each; three tool/model cycles per initial agent; 20 model calls on the ordinary path, 28 including an extra round; 40 tool calls/run; 8,192 output tokens/response; 60-second bidding/council, 60-second research, 180-second overall run. Intake is outside the run but capped at three grouped clarifications. One transient retry counts toward caps. Track tokens, elapsed time and provider cost when available. Operator sets provider spending limits before public use; stop on configured quota exhaustion. No promised hard dollar cap without provider support.

Start with gemini-3.6-flash, medium thinking; model/limits are backend settings. Truncated/invalid JSON is rejected. Increase caps only through configuration after measuring cold/warm runs; extra compute cannot bypass policy, identity, approval or time guards.

5. Real evidence and final validation

Research distinct finalist product variants after bidding, regardless of category. Merchant-selected sources are inputs, not the neutral corpus. Independently acquire manufacturer/brand facts and owner/test reports. Preflight laptops plus one non-tech product. Never fabricate quotes or transfer reviews from an unrelated product to a fictional item.

Acquisition: start an Apify Actor asynchronously; persist run ID/input digest; poll server-side at 2, 5, 10, then 10-second intervals. Verify expected run/status before reading its dataset. Webhooks can wake polling but cannot supply trusted dataset IDs or approval. Abort overdue runs where supported; late data is stale research, never advancing an obsolete request. [S4]

Bounds: eight URLs/forty excerpts per primary product variant; five primary variants including refill. Related lines use existing evidence or show not researched; do not imply full bundle coverage. Allowlisted HTTPS sources; no private-network URLs or buyer crawl targets. Deduplicate URL/content/source family. Reuse matching real evidence within 24 hours with timestamp. Cache never establishes stock, price or unit condition. Pin Actor build/input schema. [S4]

Applicability	Can support	Cannot establish
Family/model	Related owner themes and family capabilities	Exact size, material, configuration, region or unit health.
Exact variant	Explicit attributes and applicable tested performance	Another variant's fit/results or current unit condition.
Demo unit/listing	Synthetic diagnostic, stock, price and terms	Real ownership, certified refurbishment or real availability.

Evidence record: identity/applicability, URL/source ID, type, source date or unknown, retrieval time, short supporting excerpt, claim, Actor/run/build, duplicate group, ownership-label provenance and verification state. Display brief quotations and source links; minimize personal identifiers. Purge research/request data after seven days; no raw full-page retention by default.

Display: separate specifications, independent tests, owner reports and demo condition. Show positives, concerns, disagreement, count, freshness and gaps. A claimed owner is not independently verified; attribute site verified-purchase labels to that site. No artificial consensus probability, fake sentiment average or guarantee.

States: pending, complete, partial, unavailable, cached. Manufacturer-only says "owner evidence unavailable." Blocked/empty sources can use a previously collected real snapshot with original timestamp and cached label; otherwise show unavailable and source links. Advisory evidence failure does not prevent selection of a fully validated demo offer. Unknown mandatory facts still block eligibility.

Category caveats: clothing fit reports retain purchased size/measurement context; appliance performance retains region/test basis; household compatibility requires exact dimensions/interface. Mixed-variant reviews stay family-level. Product reviews cannot prove a refurbished unit's condition, seller reliability or counterfeit status.

Contradictions: authoritative facts failing a hard constraint invalidate the offer and trigger bounded refill. Conflicting authoritative facts exclude offers dependent on them until reviewed; a lone allegation/model summary cannot overwrite specs. Web text cannot grant permissions, alter policy, expose secrets or pay.

6. Data, function surface and permissions

Shared runtime-validated TypeScript contracts drive Convex, test fixtures and UI; no parallel handwritten payload definitions. Opaque IDs, integer USD cents, line quantities 1-10, UTC milliseconds; immutable records carry createdAt/schemaVersion. Pass IDs between steps; store source output separately.

Record	Minimum contract
Merchant / membership	demo=true, enabled, name; server-owned userId -> merchantId.
CategoryProfile / Product / Variant	Versioned typed fields/operators/units, clarification/compatibility/evidence rules; family identity; exact variant attributes and provenance. Unknown manufacturer IDs remain null.
Listing	Variant ID; merchant, condition/unit ID, quantity, list/floor, diagnostics, included items, version. Never collapse all sizes/colors into one stock bucket.
Policy	Merchant, immutable version, allowed terms/concessions, shipping, transit, warranty/returns, effective time.
RequestVersion / Run	Owner, constraints, private ceiling, sort, version; stage, budgets, deadline, config and latest run ID.
OfferVersion / CouncilEvent	Request/run/merchant/version, immutable lines/snapshots, terms, expiry, validity; event/claim/source references.
ResearchJob / Evidence	Identity/input digest, expected provider run/build, status; provenance/applicability.
FirmQuote / Reservation	Request/offer versions, commercial snapshot, hash, expiry; listing allocations, owner, active/locked/released/sold.
Approval / Execution / Order	Quote hash, owner/time, attempt key, mode, operation, result/unknown, receipt, mock fulfillment label.
ProcessedEvent / Audit	Logical key, payload digest/prior result; timestamps, rejections, token/cost status; no secrets/private prompts.

Public functions: authenticated and scoped

- saveDraft/confirmRequest/reviseRequest: owner; validate fields; confirmed revision immutable. Changes cancel obsolete jobs/nonlocked holds; executing requests cannot change.
- startRun/cancelRun/getBuyerView: owner; one active run/request, three active requests/session, five starts/minute. Reload resumes stored progress.
- getMerchantView/updateOwnListing/publishOwnPolicy/pauseOwnMerchant: server membership. Stock edits cannot erase sold/held allocation. Pause blocks new bids/selections, not locked-attempt reconciliation.
- selectAndReserve: owner/current offer/version; transactional recheck/hold. approveAndExecute: owner/exact quote hash/unexpired hold/idempotency key.

Internal only: dispatchAgent, validateAndStoreOffer, finishResearch, closeRound, expireHolds, executeAdapter, reconcileExecution and resetDemo. Model merchantId grants nothing. Same key + same payload returns prior result; mismatched replay fails. Every completion checks active version, expected stage/run and transition.

Auth: Convex Auth Anonymous for buyers, Password for privately provisioned operators. Create three accounts, grant memberships internally, then reject password signUp server-side. Never trust browser role/email fields. Buyers read own runs; merchants read own policies/bids and permitted council projection, never ceiling/rival floors. Separate buyer/operator browser contexts. Guest cross-device recovery is out of scope. [S2]

Errors: code, safe message, retryable, correlationId. Codes: INVALID_INPUT, FORBIDDEN, STALE_VERSION, NO_MATCH, POLICY_REJECTED, EXPIRED, STOCK_UNAVAILABLE, PROVIDER_UNAVAILABLE, LIMIT_REACHED, EXECUTION_UNKNOWN. Seller errors omit private values. Operator changes invalidate unapproved stale quotes; an executing approved snapshot completes under its frozen terms.

7. Lifecycle, reservations and execution safety

Request: draft -> bidding -> council -> researching -> results -> reserved -> executing -> complete. Alternatives: no_match, cancelled, failed. Partial/unavailable evidence annotates results. Persist state; browser/model memory is not authoritative.

On stage/overall deadline, close outstanding jobs. Continue with already validated offers and truthful partial evidence; if none exist, use no_match for confirmed inventory mismatch or retryable failed for provider failure. Late completions cannot reopen the run. Never manufacture agent output when generation failed.

Transition	Guard and outcome
Confirm/start	Owner confirms required fields; create version/run; schedule workflow. Edit invalidates old work and releases active nonlocked holds.
Bids -> results	Validate current offers; apply council/revisions; freeze current comparison pool; research/refill within limits.
Select -> reserved	Recompute current policy/listing; reject paused/unavailable/noncompliant; atomically create immutable firm quote and allocations.
Approve -> executing	Check owner/hash/versions/server time; create one approval/attempt and set execution_locked before provider calls.
Success	Convert locked allocations to demo sold exactly once; save truthful mode-specific receipt.
Definite decline/failure	Record terminal attempt; release once; return buyer to selection. Unknown is not a definite failure.

Timers: bids valid 15 minutes; no hold during research. Expired selection requires full revalidation, not automatic rejection forever: issue a renewed firm quote and display any changed terms. Firm quote/hold valid three minutes; server checks time even if cleanup is late. Successful replacement atomically releases old hold; failed replacement keeps the old valid hold.

Available = initial quantity minus sold and unexpired active or execution-locked allocations. Expired active holds do not count even before cleanup. Unique refurbished units have quantity one. Release/expiry/sale is idempotent. Never release an execution lock just because its former three-minute hold elapsed.

Payment adapter and truthful results

Default PAYMENT_MODE=simulation. No real card collection/live-money capability. Adapter input is server execution ID + approved quote; output is succeeded, declined, pending or unknown, actual operation and provider reference. Privileged simulation fixtures: success, declined, unknown_then_success. Success says "Demo order completed; no payment taken"; delivery remains fictional.

Enable Visa sandbox only after exact API/test mapping, amount/merchant semantics, duplicate protection and status lookup pass a smoke test. Disclose test-merchant mapping. An unrelated fixed test amount is not proof the chosen quote was paid. Tokenization-only says "Visa sandbox credential test"; order completion stays explicitly simulated. No invented endpoints/settlement claims. [S5]

Timeout -> EXECUTION_UNKNOWN with allocation lock retained. Reconcile the same attempt/reference; block another purchase for that request until resolved. If safe status/idempotency handling is unavailable, disable that external path and use simulation. Disable cancellation/edits during executing/unknown. Internal demo reset labels fictional-state reset; it cannot imply a provider refund.

8. Implementation Plan: architecture and setup

Runtime: React/Vite SPA -> Convex functions/Workflow -> Gemini actions and Apify jobs -> stored results -> approved quote -> adapter. Vultr serves built SPA over HTTPS with client-route fallback. Convex hosts backend; Apify hosts extraction. No permanent extra worker. [S1, S4]

Convex mutations own state/stock transactions; actions own external calls; Workflow persists orchestration/retries. Use official @google/genai behind an adapter, pin SDK/lockfile and preserve function-call parts. Source registry pins Actor builds/input mappings. Optional Product Evidence Pack Actor normalizes existing crawlers; no new crawler required. No network inside transactions. [S1-S4]

Figma and UI handoff: owners A and B

Two teammates own Figma design and the UI handoff. A owns buyer intake, requirements, progress, comparison/evidence and checkout/result; B owns merchant catalog/policy, bid/council views and shared components. This ownership split is a proposed working default. Figma work must become a connected frontend; a clickable design alone does not execute the marketplace.

Freeze the shared contracts first: CategoryProfile, RequestDraft, BuyerRunView, OfferCard, EvidencePanel, FirmQuote, ExecutionView, MerchantView and ApiError, with page-6 permission-safe fields. Backend owns totals, eligibility, rank, timestamps, modes and allowedActions; UI renders them. Profile fields drive forms/spec rows, never hardcoded laptop columns. Fixture payloads cover all four category families. Figma may supply components/tokens; server data remains authoritative.

Design/handoff must cover responsive layouts, keyboard/focus, form validation, loading/empty/error, stale offers, partial evidence, expired hold, payment pending/unknown and success/decline. Persist IDs across reload; use subscriptions for progress. Council detail shows claims/outcomes, not private reasoning. A/B own frontend wiring by default, C/D supply tested contracts and adapters; no unassigned conversion from design to code.

Figma tracks: supplied and current event listings include Best Use Case of Figma, Best Design, Most Creative, plus two Figma Agent categories. Preserve the design file, prototype, iterations and live implementation evidence. Agent categories require actual demonstrated Agent use and rule confirmation; ordinary Figma use does not establish that. Figma is a development/design track, not an added runtime dependency. [S6]

Configuration

Browser: public VITE_CONVEX_URL/demo labels only; VITE variables are public. Backend: GEMINI_API_KEY, GEMINI_MODEL, APIFY_TOKEN, source registry, PAYMENT_MODE, optional Visa test credentials/mapping and Auth signing settings. Server constants own limits/TTLs/region. Keep credentials and provisioning gates out of client/public source. [S1-S4]

Start from the official Convex React/Vite + Auth scaffold. Seed catalog/policies and privately provision operators; disable public password signup. Default inventory mode is mock. Never assume credits or generate credentials in source.

Release sequence

1. Freeze contracts/profile definitions and Figma states; seed fixtures/auth; prove owner and merchant isolation.

2. Test real Gemini tools and Apify sources for a laptop and non-tech item; pin working versions. Failures become honest cache/unavailable states.

3. Connect one full offer -> hold -> approval -> simulation path, then council/evidence/refill; exercise each category without a backend fork.

4. Configure deployment Auth/Convex separately; build dist for HTTPS Vultr hosting with index.html fallback. Vite preview is not production. Test nested refresh/fresh sessions.

5. Run page-9 cases, gated Visa test and submission proof. No deployment/spend/publication performed by this planning task.

Use the laptop pitch plus a short non-tech run; clearly distinguish synthetic fixtures, live/cached evidence and simulation.

9. Impact, acceptance tests and four-person delivery

Measure time to a qualified offer, comparison effort, valid/distinct offers, policy rejections, model/tool calls, cold/cached latency, source coverage, changed checkout terms and duplicate attempts. Merchant contribution is fictional until real data exists. Budget minus price is not measured savings; comparisons need equivalent delivered terms.

Test group	Required expected result
Intake / categories	Broad/exact input confirms a profile. Windows/16GB/512GB rejects P6. Cotton 100% + DemoWear M rejects C2; width <=600 mm rejects H2; US 120 V/60 Hz plug rejects A2. No laptop fields required on clothing. Unknown category stays draft.
Policy	Below-floor, wrong SKU, false warranty, incompatible accessory, bad cents/quantity or unknown hard fact rejected despite valid JSON. One repair or abstention.
Privacy	Buyer A cannot read B; merchant A cannot read/edit B; forged roles fail; ceiling/floors absent from opponent prompts, public data and logs.
Council / rank	Three actual isolated tool-using contexts; bounded rounds; rhetoric changes no rank; same validated snapshot sorts identically; extra round logs a material trigger.
Evidence	Wrong configuration/size/region not exact evidence; unknown health fails threshold; synthetic C/H/A fixtures get no borrowed reviews. Laptop and genuine non-tech source retrieval tested; blocked/injected pages stay safe.
Version / refill	Edits make old completion stale; factual invalidation refills once within limits; fewer results truthful; no fabricated review/restart.
Inventory	Last unit: one hold wins; delayed cleanup cannot permit expired approval; duplicate release restores nothing twice; failed selection swap preserves valid hold.
Execution	Double click: one attempt/order; expiry preserves execution lock; unknown_then_success reconciles once; decline releases once; tokenization never claims charge.
Recovery / UX	Reload resumes; timeout produces partial result/retryable failure; nested refresh works; modes visible. All Figma handoff states connect to real typed responses; keyboard completion works.

Milestones / owners

Milestone	Work split and exit gate
M0: foundation	A/B Figma states and UI contracts. C category/catalog/policy/auth contracts + model tool test. D state/holds/payment + source access test. Agree schemas before parallel UI work.
M1: thin path	A buyer UI + B merchant UI wired to C valid offer + D hold/approval/simulation. Reject unauthorized transitions.
M2: differentiation	C merchant agents/council/profile validation. D evidence/identity/refill and Workflow. A/B progress/comparison/council/evidence screens; exercise laptop, clothing, household and appliance fixtures.
M3: hardening	C policy/privacy/adversarial tests; D lock/reconciliation, Visa gate and source failures; A/B all error/expiry/accessibility states.
M4: submission	A/B Figma and demo evidence; C/D integration proof. All repeat runs, video/image/repo, truthful modes and category eligibility.

With two backend builders, prioritize the shared flow and category tests. If time runs short, cut optional integrations/Actor packaging before category support; retain a laptop and one non-tech end-to-end demo. Candidates: Visa, Apify, Convex, Gemini API, genuine Vultr hosting and Figma; Best Actor/Figma Agent conditional. Fetch AI excluded by UI choice; Spacetime replaces Convex. Elastic/Factory/Cartesia deferred. Compare agents against plain catalog filtering.

10. External gates, sources and readiness verdict

Dependency	Status / safe behavior
Merchant authority	Resolved for prototype: fictional merchants/stock approved and labeled. No live seller permission needed for this demo.
Provider accounts	Convex/Gemini/Apify access and quotas untested. Actual use must be demonstrated; fixtures/contracts can be built while obtaining keys.
Source coverage	Exact Actor/source pairs need testing for laptop and genuine non-tech item. Real cache/unavailable is valid; synthetic reviews prohibited. New category profiles need field/provenance tests.
Visa	Operation/access unknown; simulation reliable default. Enable only a tested, accurately labeled capability.
Figma / eligibility	A/B own design/UI. Confirm Figma access, exact judging proof, Agent access if targeted, category/award limits and Best Actor publication rules. No prize eligibility guaranteed.
Deployment	Confirm sponsor accounts/credits and hosting access; no spend/public deployment performed here.

Primary implementation references

- S1: Vite deployment, environment, Convex actions, Workflow, scheduler.
- S2: Convex Auth setup, Anonymous, Password, authorization. No client-controlled role assignment.
- S3: Gemini 3.6 Flash, function calling, structured outputs. Shape does not establish truth.
- S4: Apify async runs, running Actors, webhooks. Source access/cost/output need testing.
- S5: Visa Intelligent Commerce, quick start. Tokenization alone is not checkout.
- S6: Event requirements, Fetch AI hackpack. ASI:One requirement conflicts with chosen UI; award stacking unconfirmed.
- S7: Apple MacBook Air M2 2022 US specs.
- S8: Apple MacBook Air 13-inch M3 2024 US specs.
- S9: Apple MacBook Air 13-inch M4 2025 US specs.
- S10: Lenovo US 21ML0053US. Specifications, not availability.

Verification: earlier independent reviews covered state/payment, agents/evidence and stack/auth. This revision additionally checks category identity, constraints, fixtures, ownership and Figma handoff consistency. Primary docs/Context7 informed integrations; current event Figma categories checked. These are proposed contracts, not passing application tests. No model/Actor/auth/reservation/payment execution tested; layout checked separately.

Open questions for sponsor access, not a design freeze: which Visa sandbox operation is available; which Actor/source pairs work; which Figma/Agent submission evidence and award combinations are accepted? Assign these to M0; implementation defaults above can proceed while these are resolved.

TLDR: v3 specifies a multi-category marketplace, with laptops as one demo, disclosed mock commerce, real applicable evidence and two Figma/UI owners. Category profiles, contracts, state/permission guards, compute limits and acceptance outcomes support implementation. Begin M0 contracts/source tests; external access, eligibility and runtime performance remain unverified.